

Industrial Metaverse - Virtual Reality Platform for Rapid Lattice Prototyping

Jason Chen, University of Pennsylvania

Project Overview

This research project proposes to develop an immersive virtual reality platform for lattice-structured products for additive manufacturing that unifies the following capabilities:

- Parameter tuning & product iteration with live visual & haptic feedback
- Haptic gloves give users estimate of product's mechanical properties
- Various interactions allow users to interact with and inspect products

Research

Initial Roadblocks

From previous research, interaction did not feel natural. Haptic gloves were janky, and there was no visual feedback. Also, the lattice took several minutes for computation within Grasshopper, and 10 minutes for importing and decimating (simplifying geometry) using Unity. Since STL files are complex and potentially corrupted, they took even longer to import.

File Type	File Size	Vertex Count	Average Import Time	Average Import Time + Decimate
FBX	10 MB	2 million	8 seconds	1 minute
GLTF binary	17 MB	2 million	8-10 seconds	1 minute
STL	36 MB	2 million	30-45 seconds	5 minutes
OBJ	80 MB	2 million	8-10 seconds	1 minute
FBX	60 MB	8 million	1 minute	15-20 minutes
GLTF binary	174 MB	8 million	1 minute	15-20 minutes
STL	373 MB	8 million	Crashed	Crashed
OBJ	850 MB	8 million	1 minute	20 minutes

Figure 1: Average import times for each file size, complexity, and method using Unity Asset Transformer

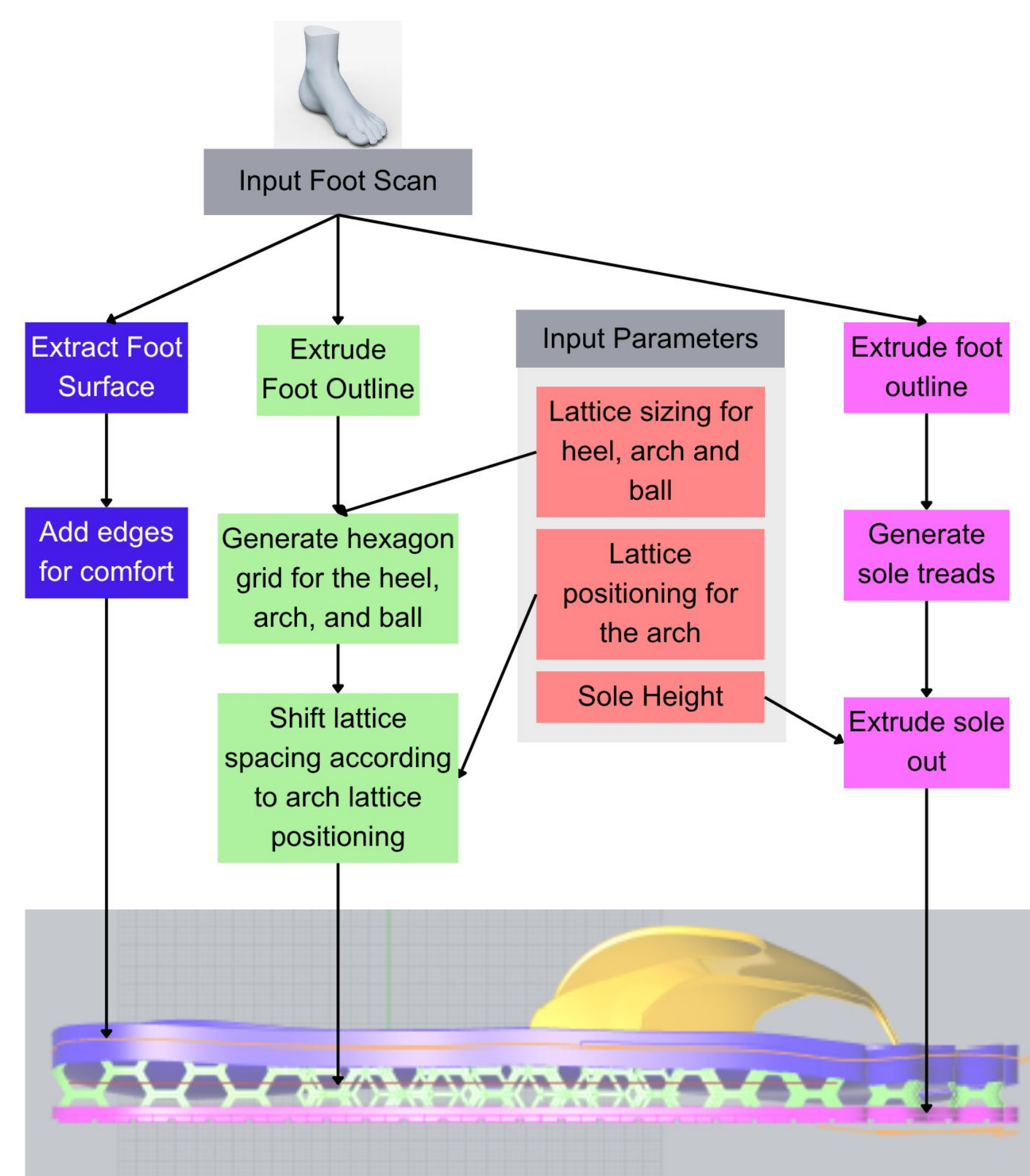


Figure 2: Grasshopper sandal design configurator workflow, and the resulting sandal generated

Sandal Configurator

A design configurator using Grasshopper made by Jenjira Burapharat for customizable sandals was chosen. After a foot scan is inputted, the plantar surface is extracted, used for the insole and outsole, while a hexagonal grid is extruded to fill the midsole. When changing parameters related to the sole or straps, Grasshopper would take an average of 1 minute of computation time. However, when changing parameters related to the lattice structures, Grasshopper would take an average of 2 minutes for computation.

Platform Design

To transport data, the platform implements a bidirectional data loop between Unity and Grasshopper to enable interactive parametric modeling. User interactions in Unity, like modifying lattice size parameters, trigger a script that writes the new data to a JSON file, causing the parametric definition to be recomputed in Grasshopper after it's script reads the JSON file. The updated geometry is then triangulated, serialized into a binary format with only vertex and triangle data, and written to disk. Unity continuously monitors this output using a server, and upon changes, it deserializes the mesh, and replaces the existing geometry in the VR scene with the updated model. This workflow allows import times to be **near instant**, a significant improvement from Unity's built in importing system. Now, the only bottleneck is 1-3 minutes of Grasshopper computation.

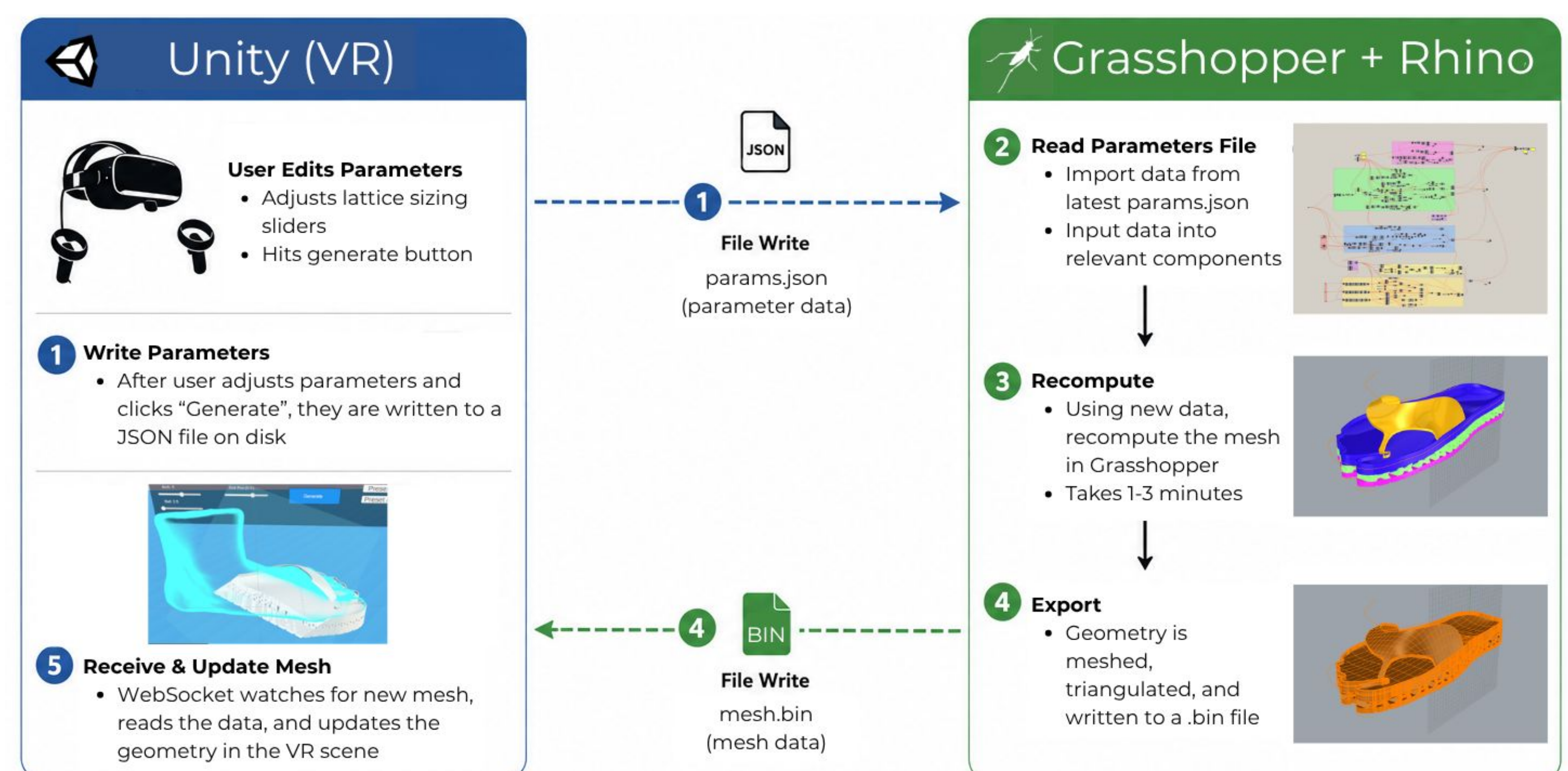


Figure 3: Unity to Grasshopper data flow design.

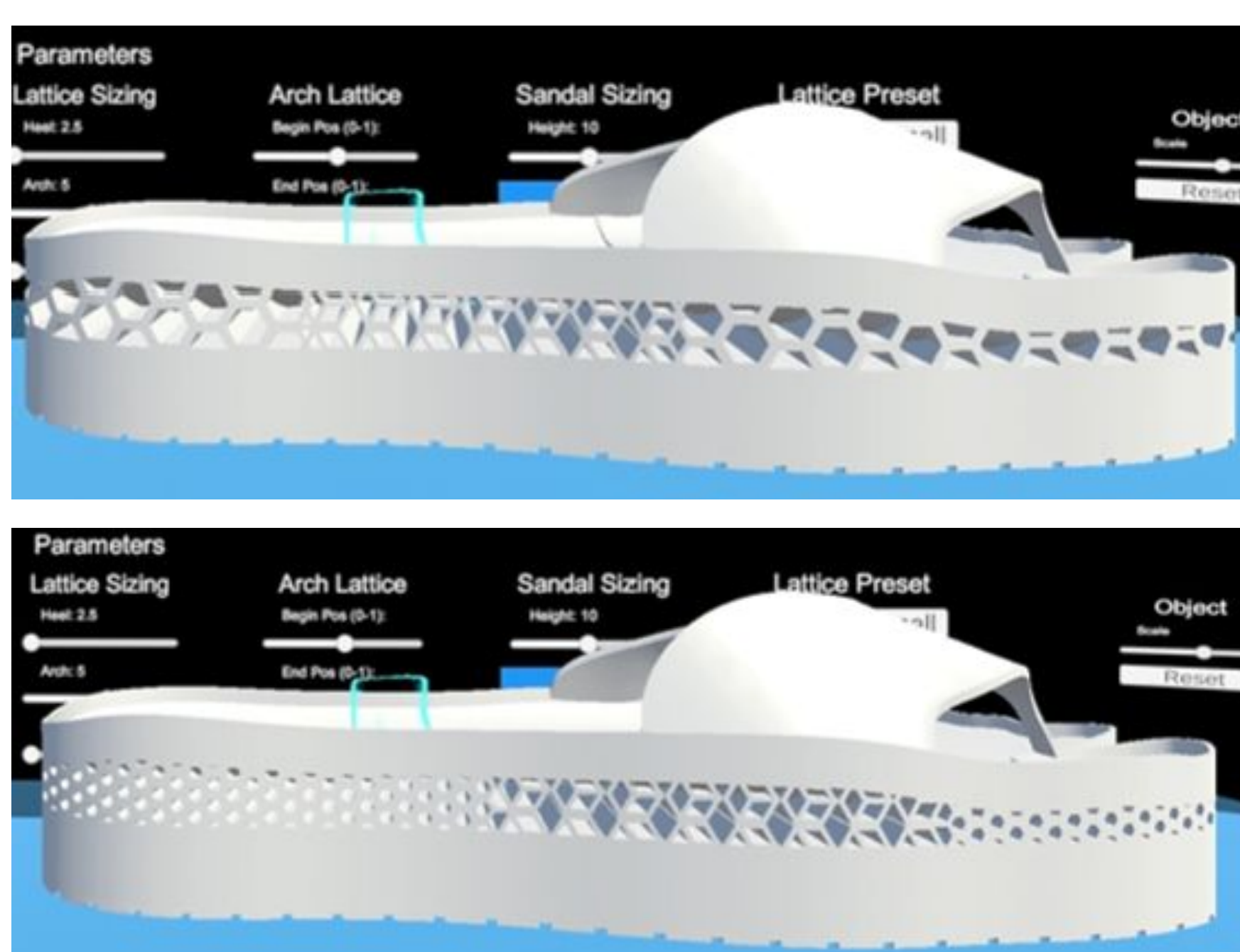


Figure 4: Before and after of sandal with heel and ball lattice size reduced, while the arch lattice position is shifted forward

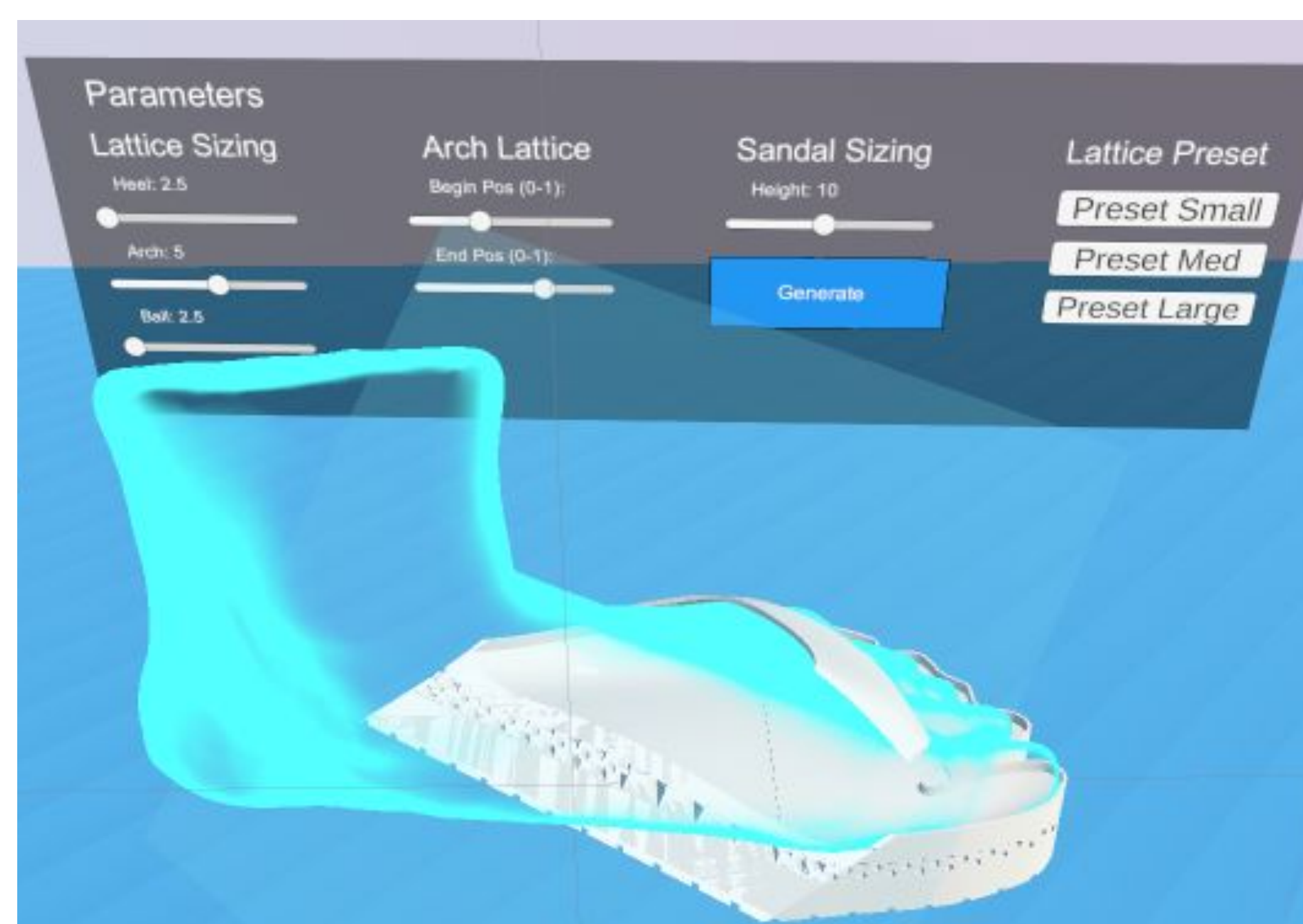


Figure 5: Platform interface, the foot scan with the demo sandal displayed in cross section.

Interface

The platform was equipped with many different interactions. Users are able to toggle on and off Augmented Reality (AR). For the product itself, several parameters are exposed for easy editing. Options like the radius of the lattice cell, positioning, thickness, etc. can be edited, and the result shown soon after they generate the product. Finally, to interact with the product, users are able to scale it up or down, look at cross sections, and feel the hardness of the product. Currently, perceived hardness is based on calculating the density of the lattice.

Conclusion

Evaluation

After user testing it became evident that the interactive platform must update instantly, as average computation times of many minutes break immersion. Scaling and cross-section tools were well received and gave users freedom. The current haptic-glove and headset setup prevents users from moving around the scene, and the perceived hardness of the lattice is not yet accurate to real physical properties

Next Steps

1. Computation times must be made faster by breaking the Grasshopper configurator into smaller parts, and reorganized. Rhino.Compute could be used to outsource computation power to servers.
2. User interaction should be richer and more customizable: swapping foot scans, adjusting sandal shape/curvature, etc. Rhino.Inside embedded in Unity could allow users to do more like adjusting curves & points
3. Current haptic gloves only vibrate on four fingers and only at the tips, and the estimated hardness is not realistic right now.